

GDL 60 Configuration w/ Remote Aircraft Status

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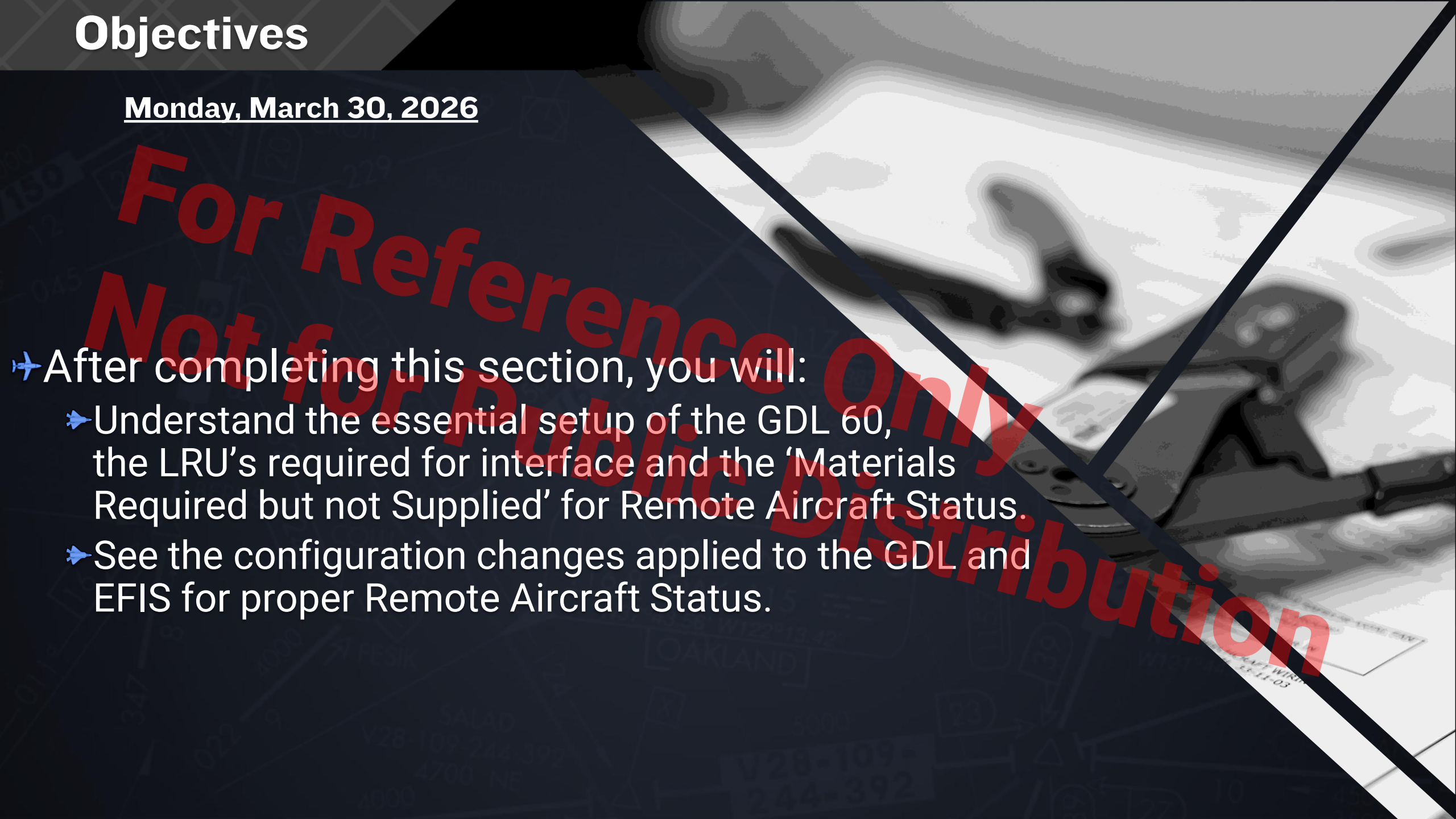
Objectives

Monday, March 30, 2026

- ✈ GDL 60 Essentials
 - ✈ Setting up the subscription
 - ✈ General purpose and RAS necessities
- ✈ Customer States!
 - ✈ Scenarios and lessons learned

Objectives

Monday, March 30, 2026

- 
- ✈ After completing this section, you will:
 - ✈ Understand the essential setup of the GDL 60, the LRU's required for interface and the 'Materials Required but not Supplied' for Remote Aircraft Status.
 - ✈ See the configuration changes applied to the GDL and EFIS for proper Remote Aircraft Status.

- PlaneSync
 - ▶ “Input free” Database Downloads
 - ▶ Over WIFI
 - ▶ Over LTE
- Higher Speed HSDB
 - ▶ 100 MBPS
- Internal Storage for Downloaded Databases
- Connex Features
 - ▶ Crew

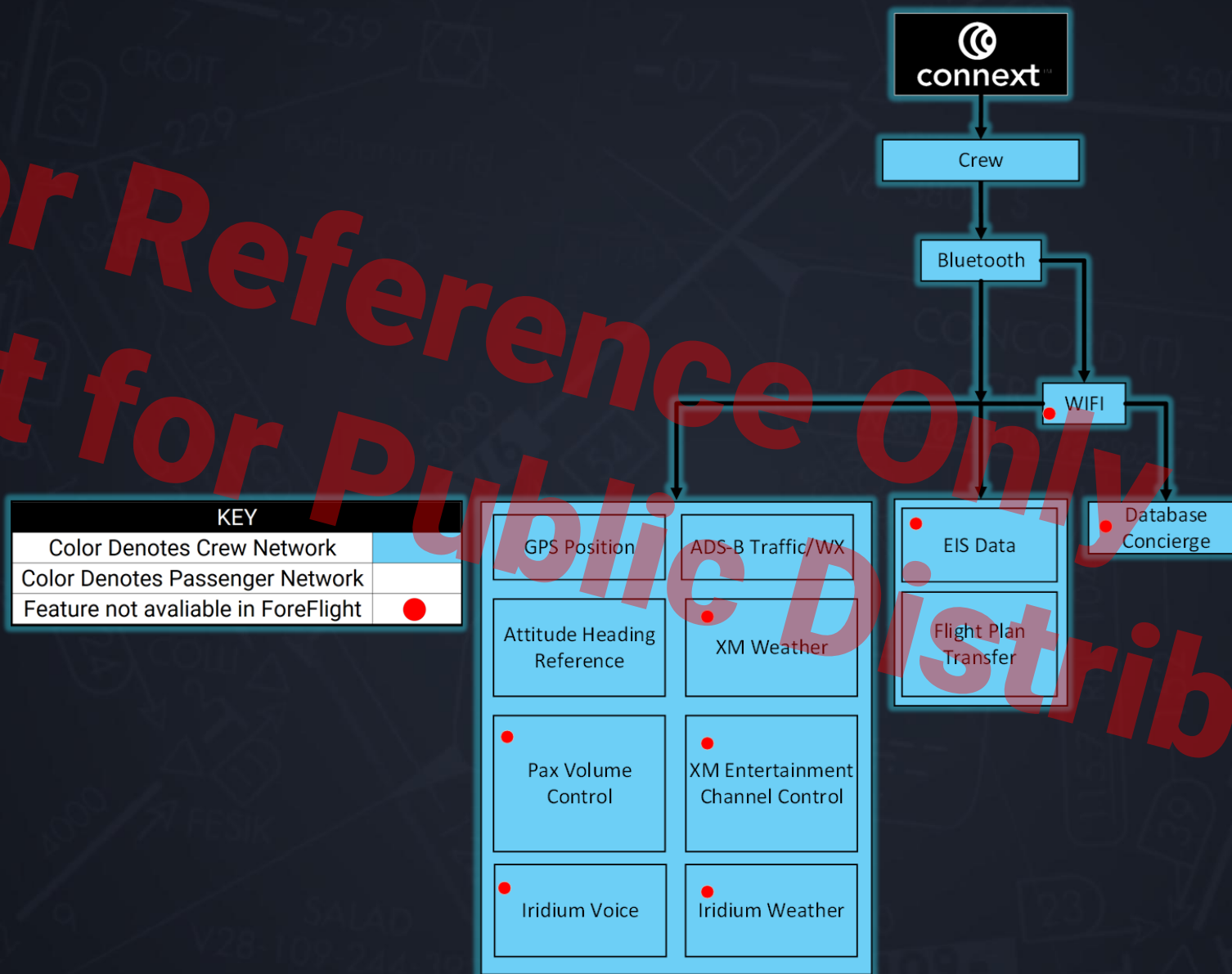


✈️ Remove the flight stream 510
when a GDL 60 is installed



- ✈ Remove the flight stream 510 when a GDL 60 is installed







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- Manage Subscriptions
- Update Schedule

SERVICES

- Connected-Cockpit Services

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Not sure what to do next? The videos below can help you get up to speed.

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Support 0

Overview Devices Buy Databases Update Schedule Warranty Connected-Cockpit Services Transfer Aircraft



Connected-Cockpit Services

Account: **Nick Wright**

Aircraft Services Account & Invoices Plans & Pricing

Aircraft

How can I add an aircraft?

Aircraft	Make/Model	Serial Number	System	Active Services	Current Charges
No Rows To Show					

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Using Connext services to access the Iridium Certus® satellite network, the GSR 56C datalink can bring on-demand global weather information - as well as onboard text/voice communications, aircraft position tracking and more - to the cockpit experience. A satellite service subscription is required, and a variety of plans are available.

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GDL 60 Datalink Services

To simplify managing an aircraft, the GDL 60 datalink technology can provide remote access to your aircraft's status via the Garmin Pilot app, while it automates tasks such as avionics database updates using LTE cellular or Wi-Fi network connectivity. View the coverage area for LTE plans.

Select Your Aircraft to Sign Up for a GDL 60 Service:

N1776 (Training TBM) ▼

Cancel Continue

LTE North America

Automatic Database Updates:
Automatic database updates via LTE within the United States, Canada, or Mexico and via Wi-Fi worldwide^{4,5}

Remote Aircraft Status:
Remote aircraft status via D2 Mach 1 Smartwatch and the Garmin Pilot app⁶

\$30 / Month⁷

Pricing starts at \$30/month. Roaming fees apply.

LTE Worldwide

Automatic Database Updates:
Automatic database updates via LTE or Wi-Fi worldwide^{4,5}

Remote Aircraft Status:
Remote aircraft status via D2 Mach 1 Smartwatch and the Garmin Pilot app⁶

\$45 / Month

Pricing starts at \$45/month.

Wi-Fi Only

Automatic Database Updates:
Automatic database updates via Wi-Fi worldwide⁵

\$10 / Month

Pricing starts at \$10/month.

Aircraft Maintenance Reports - LTE and Wi-Fi

Transmit Reports:
Transmit aircraft maintenance (CMC) reports via LTE or Wi-Fi worldwide⁴

\$50 / Month

Pricing starts at \$50/month.

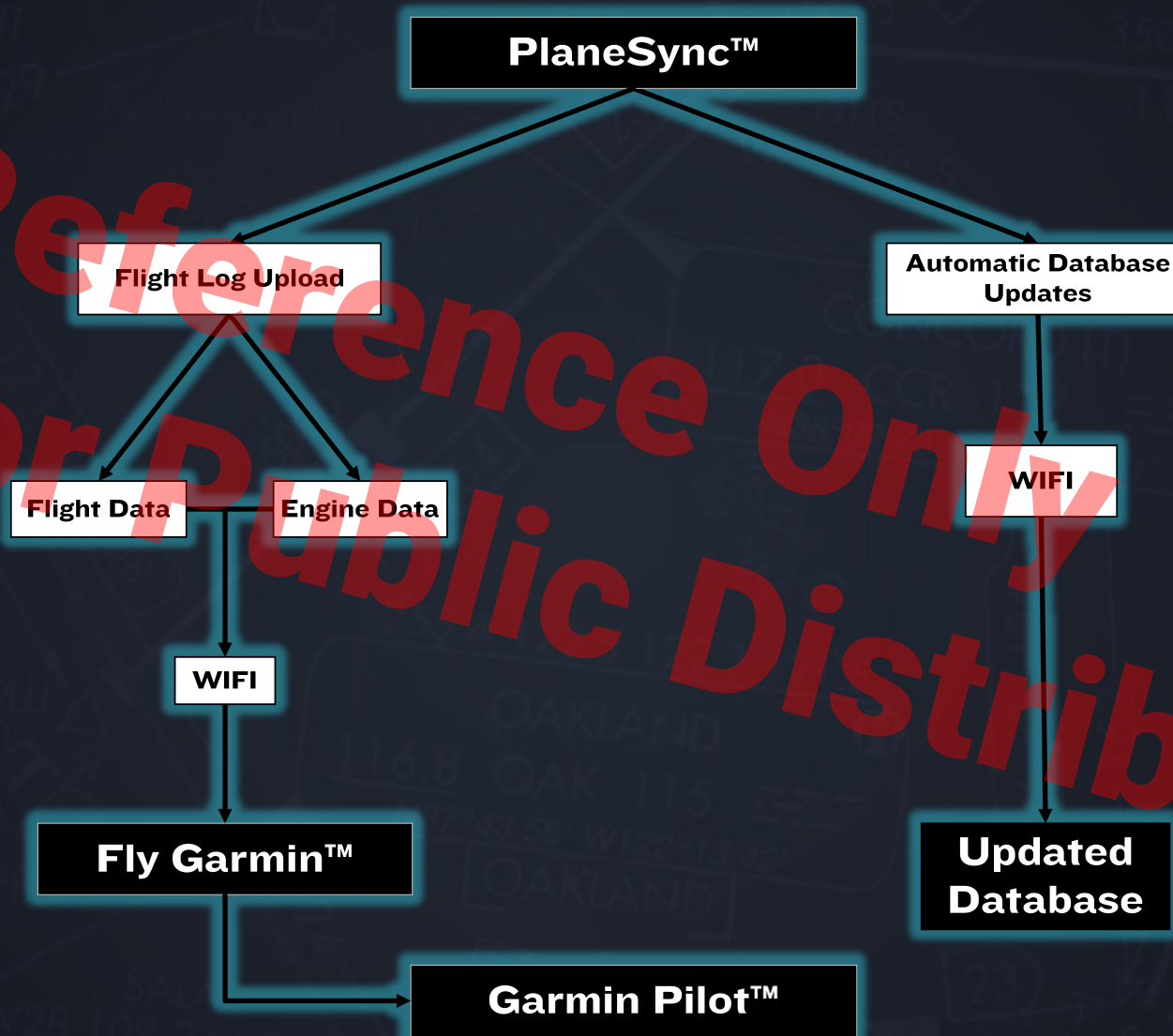
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HAVE QUESTIONS?



Minimum equipment

- GTN Xi (X1)
 - GDL 60 (X1)
 - ✈ GA60 (X1) and 2.4 GHZ (X2)
 - ✈ GA61 (X1) and 2.4 GHZ (X1)
- One or the other

Minimum Wiring

- Power
- Ground
- HSDB
- Fuse

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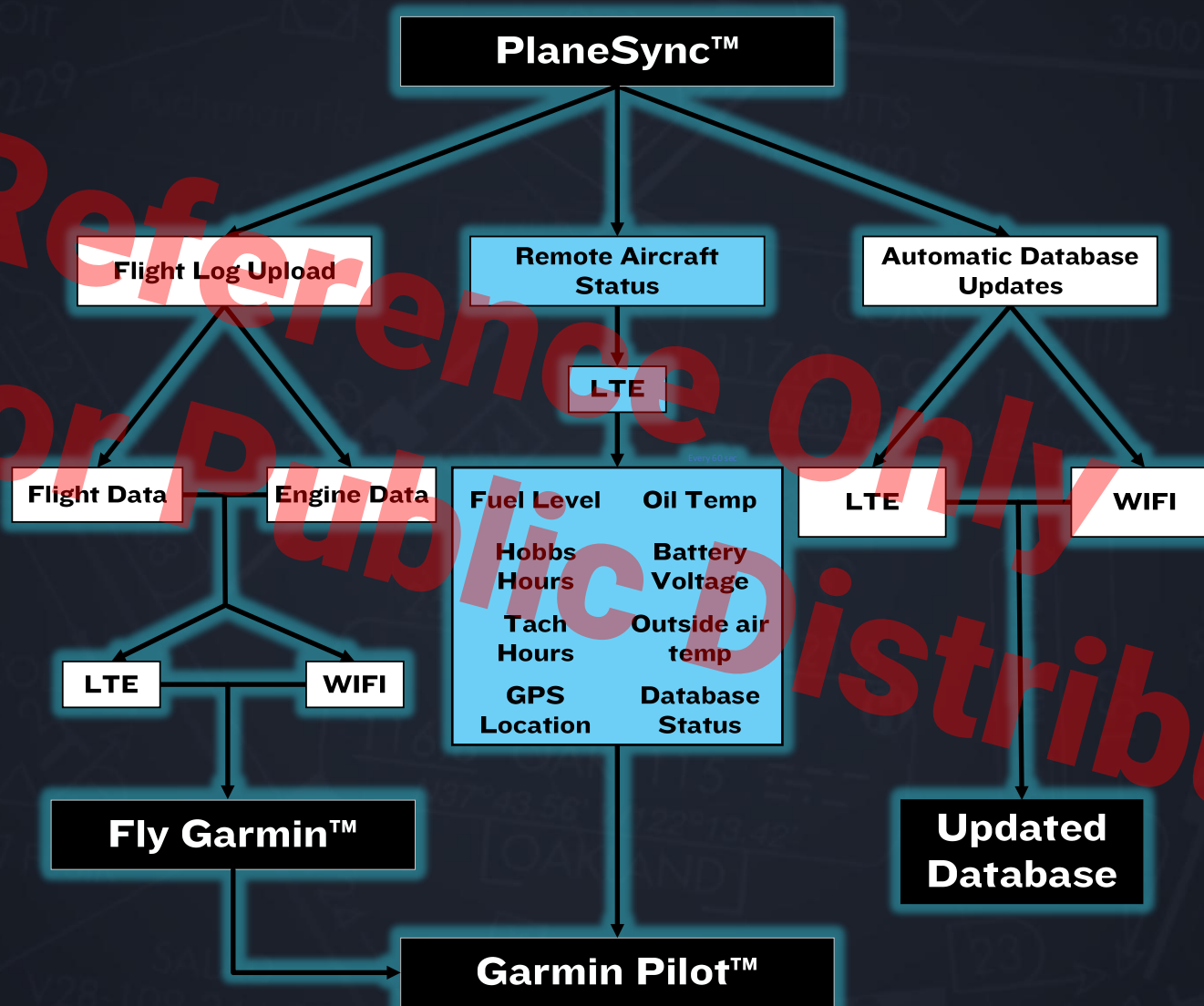
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HAVE QUESTIONS?



- LTE can only function where the signal is present there are a few things that can interfere with LTE signal

- ✦ Distance from source

- ✦ Typically cell signal can work about 45 miles from the tower
- ✦ Not all cell towers are Garmin LTE towers

- ✦ Signal Interference

- ✦ Landscape features (buildings, Mountains etc.)
- ✦ Faraday Cages



- A Faraday cage is a protective enclosure that prevents certain types of Electromagnetic radiation(EMR).
- It is essentially a container, or a shield that blocks EMR
 - EMR includes radio waves and microwaves (I.E. Cell signal)
- Why do I need this
 - A faraday cage is constructed of a conductive material (like metal)
 - What are most hangars made out of?
 - This can have varying effects on LTE signal
 - ✦ Can be remedied with an LTE repeater



Minimum equipment

- ✈ GTN Xi (X1)
 - ✈ GDL 60 (X1)
 - ✈ GA60 (X1) and 2.4 GHZ (X2)
 - ✈ GA61 (X1) and 2.4 GHZ (X1)
- One or the other

Minimum Wiring

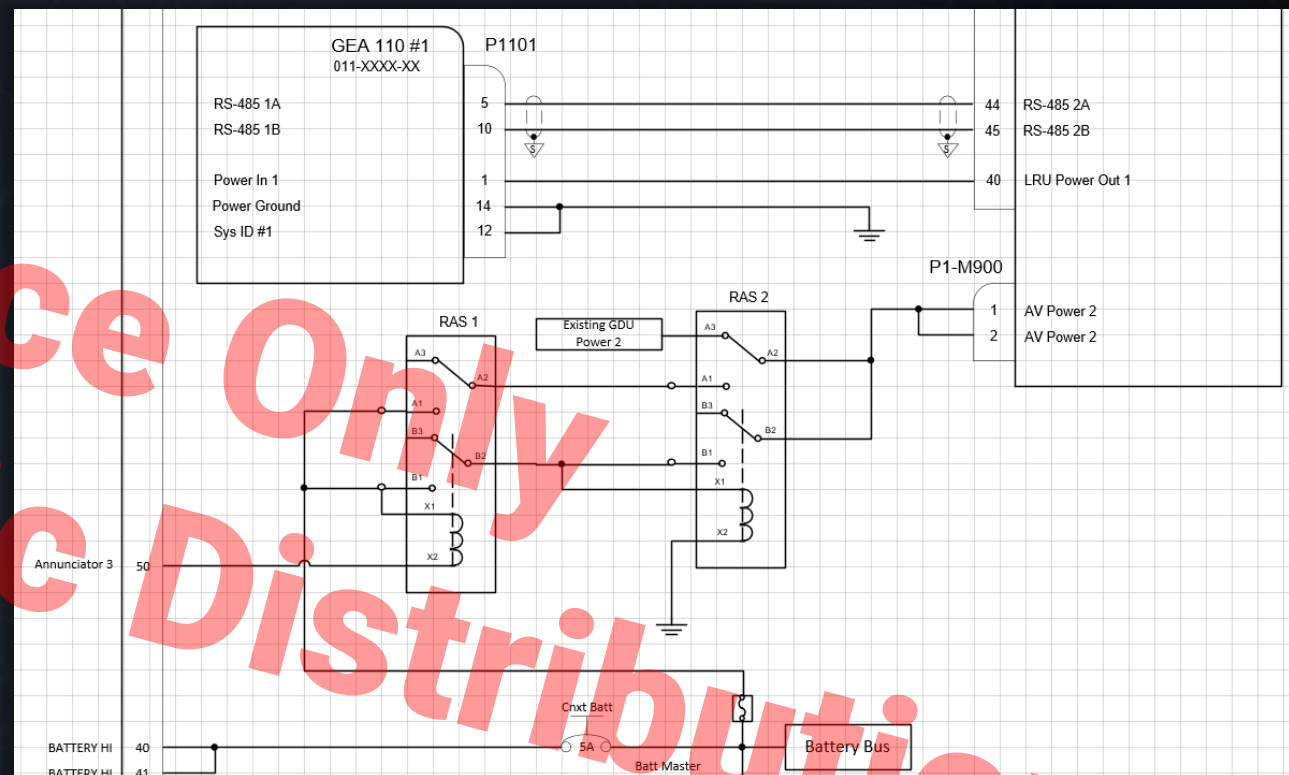
- ✈ Power
- ✈ Ground
- ✈ HSDB
- ✈ 2 Relays
- ✈ Fuse

Choose your own adventure

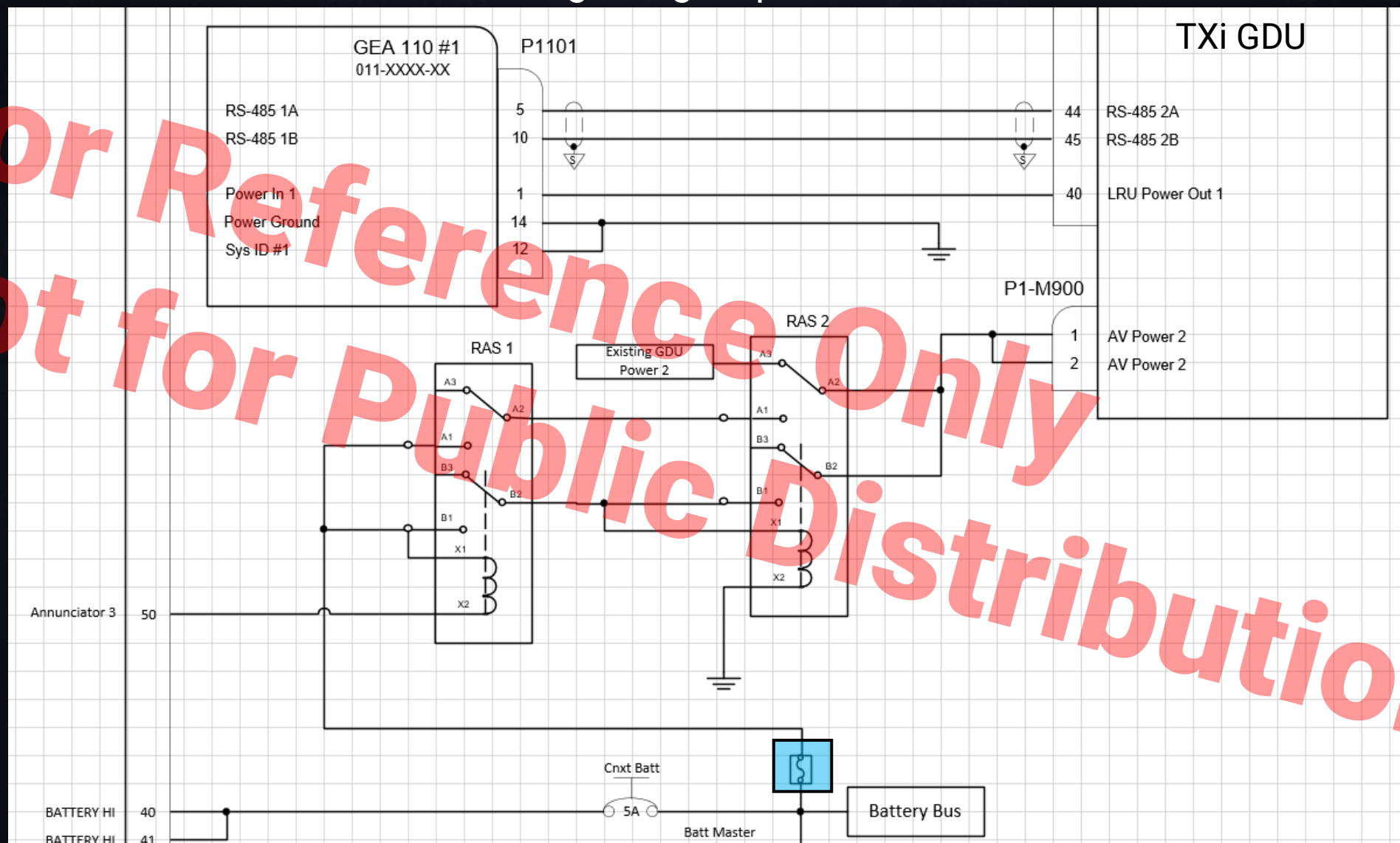
- ✿ We gave guidelines and many examples because this can change depending on components and configuration on airframe



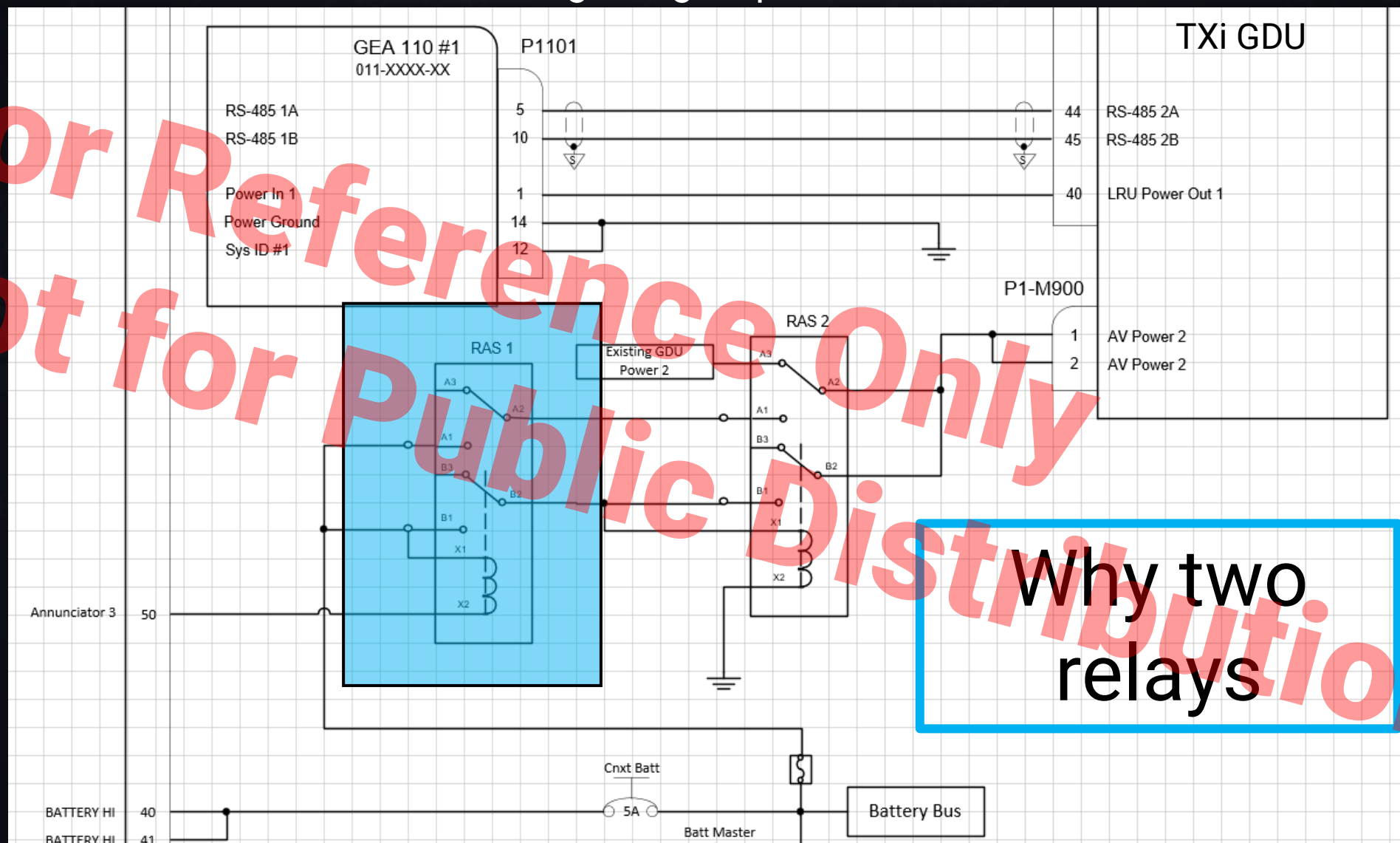
✈ The GDL 60 utilizes ship battery or a standalone battery to provide power to necessary LRU's and sensors during a RAS event



Single engine piston



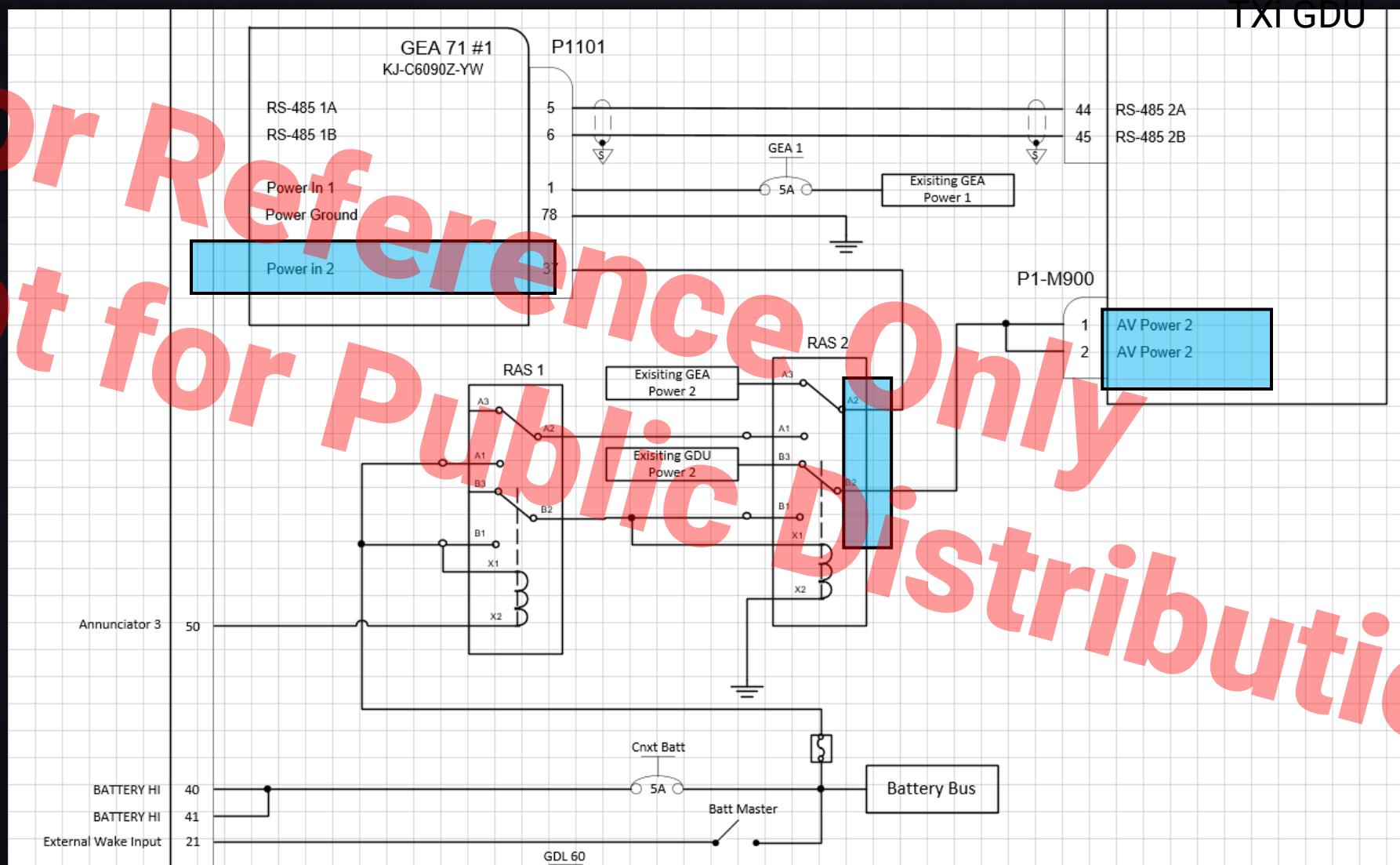
Single engine piston



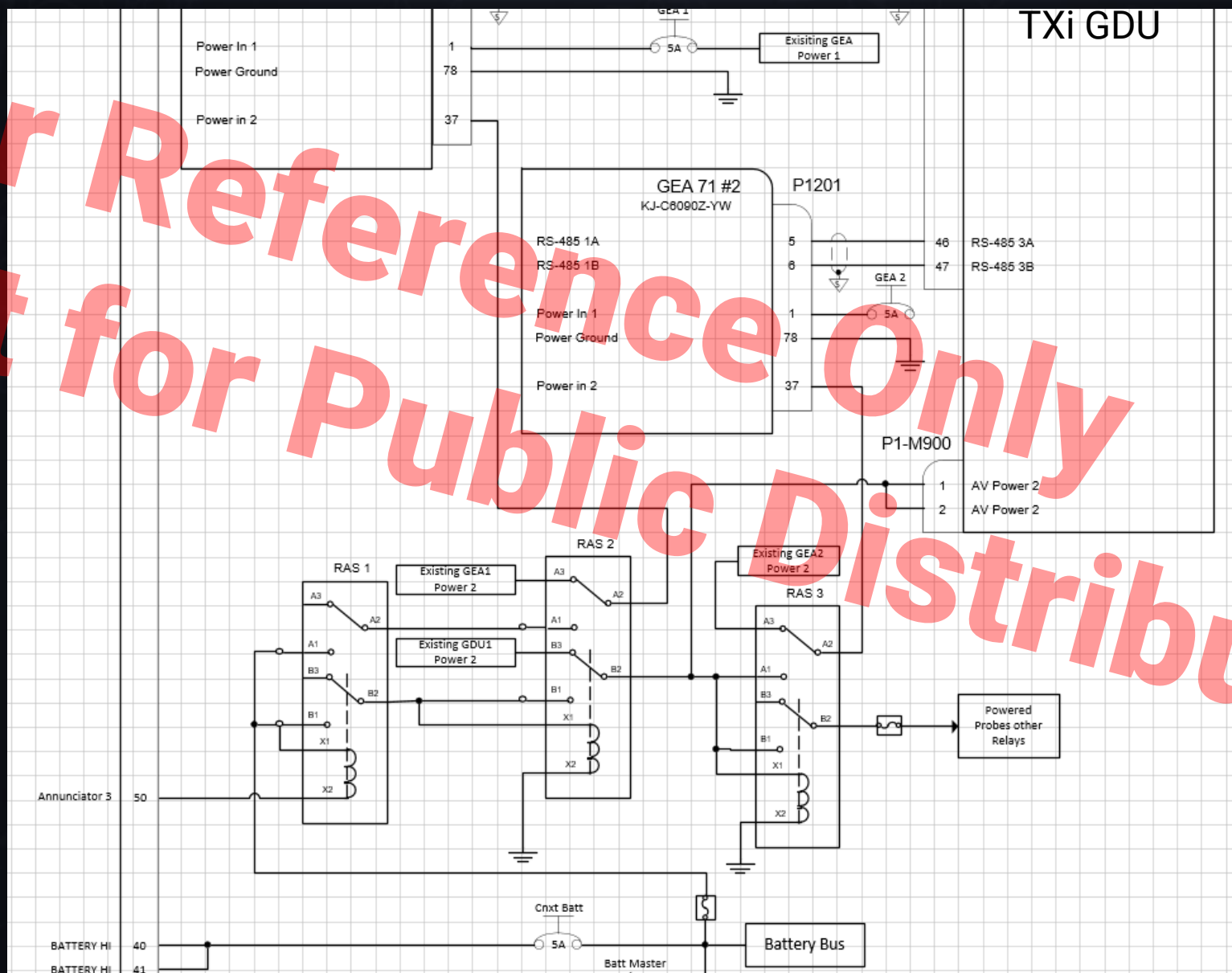
✈ Sec. 23.1361 — Master switch arrangement.

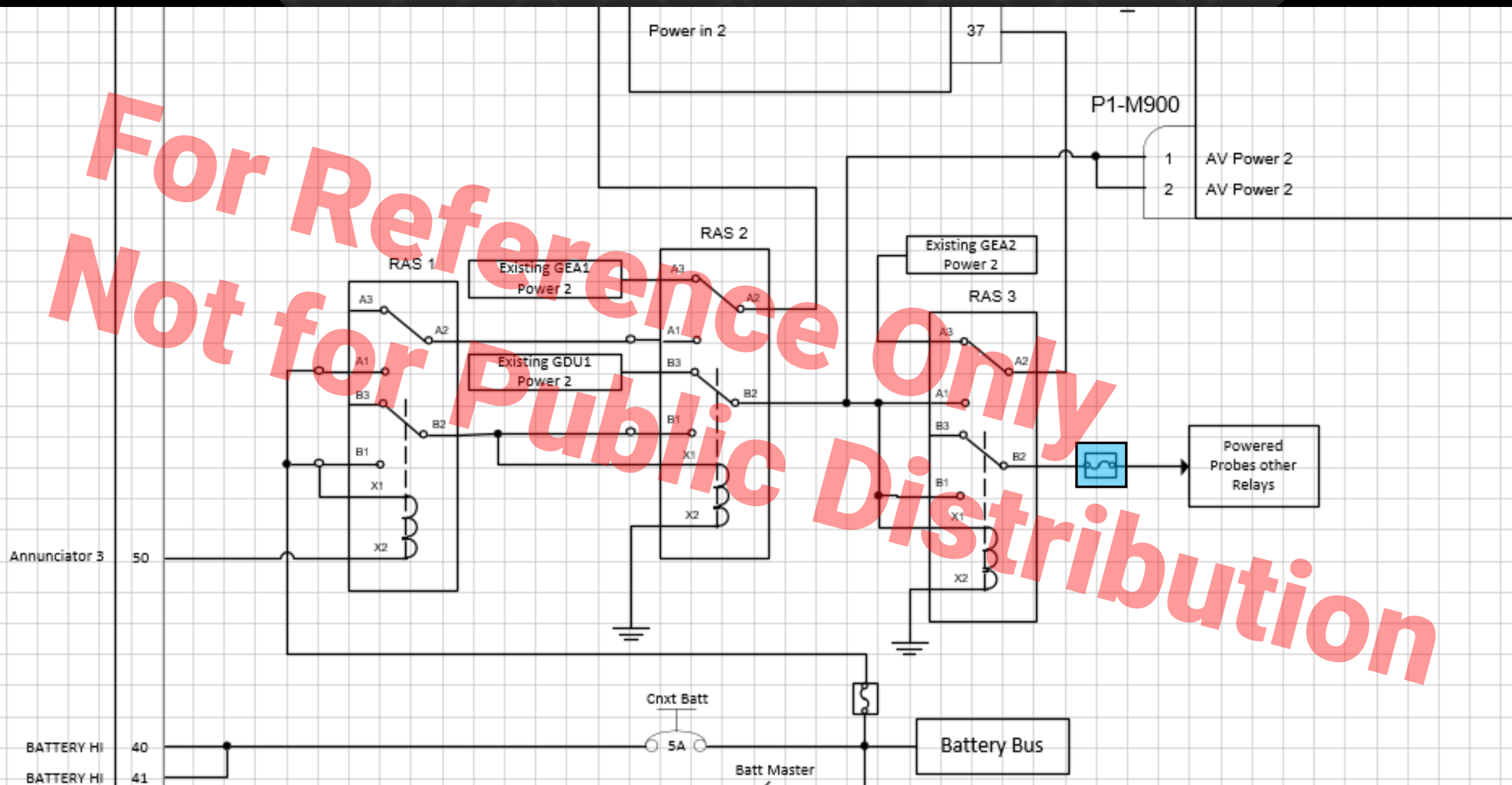
- ✈ (b) Load circuits may be connected so that they remain energized when the master switch is open, if the circuits are isolated, or physically shielded, to prevent their igniting flammable fluids or vapors that might be liberated by the leakage or rupture of any flammable fluid system; and

Single engine turbine



Multi engine turbine





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Customer States

Real world scenario

- ✈ This is the Customer aircraft 3 TXi displays a GI 275 standby and Dual GTN Xi's
- ✈ The customer Complained that the GDL 60 was doing automatic databases. (very happy with that) however its not offloading flight data logs, AND RAS was intermittent...



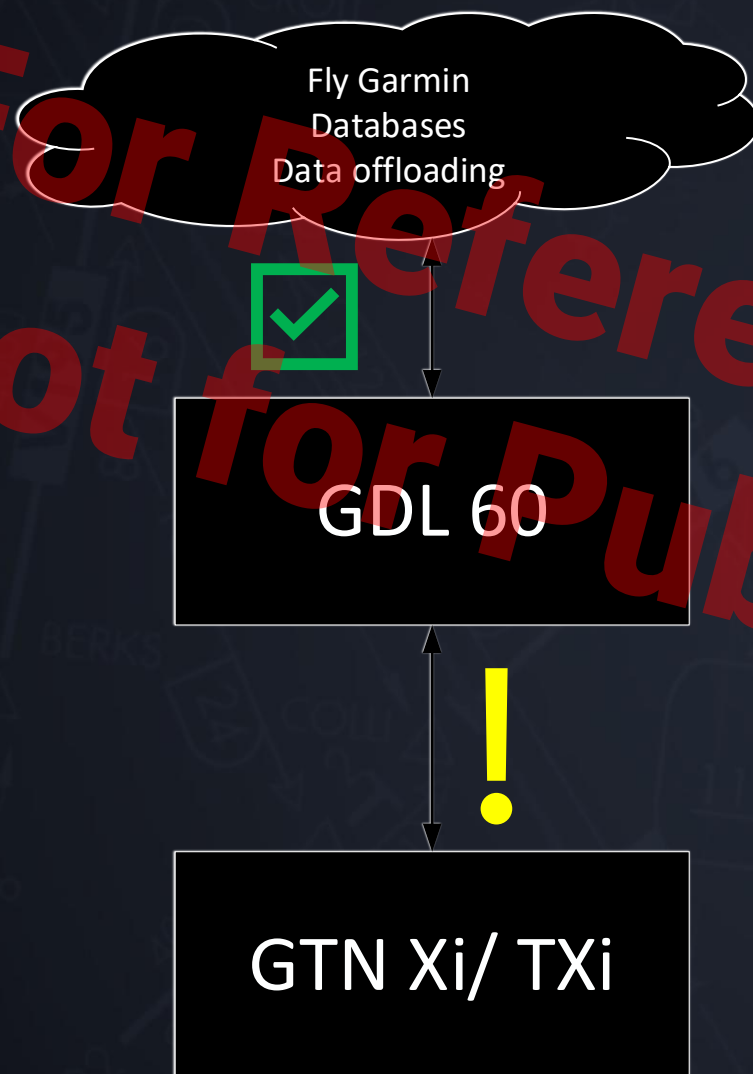
✈️ The GDL 60 was getting databases.

✈️ This tells us that the GDL 60 is connected to the server and receiving info.

✈️ That means we have a good subscription, and it is communicating with the Garmin servers

✈️ We know that that connection is good!





✈ The customer Complained that the GDL 60 was doing automatic databases. (very happy with that) however is wasn't offloading flight data logs.

✈ So this points at the EFIS Configurations since everything works... sort of.

- ✈ We know the GDL 60 Communicates with the system over HSDB
- ✈ In this case directly with the EIS TXi Display
- ✈ The installer also elected to put a secondary HSDB from the GDL 60 to the GTN Xi



✈️ Let's look at the configs on the TXi Displays

✈️ GDU 1

✈️ This display looks correct

GPS 1	Interface	GTN 7XX
GPS 2	Interface	GTN 6XX
NAV 1	Interface	GTN 750
NAV 2	Interface	GTN 650
Standby Instrument	Interface	GI 275
ADS-B	Interface	GTX 345
Traffic	Interface	None
GDL 60	Interface	Other LRU
GDL 69	Interface	GDL 69A
	Antenna Gain	22.00
	Cable Loss	3.0
	Calc Atten	5.0
	Automatic Muting	Off
	Mute Speed	0 KT
GSR 56	Interface	None

	RS-485 RX/TX	Port 3
	Part Number	011-03454-01
GPS 1	Interface	None
GPS 2	Interface	None
NAV 1	Interface	None
NAV 2	Interface	None
Radar Altimeter	Interface	None
Backup Battery	Interface	None
Standby Instrument	Interface	None
GDL 60	Interface	HSDB Port 2

Standby Instrument	Interface	None
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✈️ Let's look at the configs on the TXi Displays

✈️ GDU 2

✈️ Here we see issues:

✈️ The GPS Isn't configured at all,
The GDL 60 is configured, but the
GI 275 is not.

✈️ This means while it can receive some data from the HSDB, it doesn't look for these LRU's to send and receive data

✈️ You will also notice this is the EIS unit the GDL 60 is directly connected too

✈️ Let's look at the configs on the TXi Displays

✈️ GDU 3

✈️ This Unit is configured correctly

Part Number		011-03454-01
GPS 1	Interface	GTN 7XX
GPS 2	Interface	GTN 6XX
NAV 1	Interface	GTN 750
NAV 2	Interface	GTN 650
ADF 1	Interface	None
ADF 2	Interface	None
Radar Altimeter	Interface	None
Standby Instrument	Interface	GI 275
ADS-B	Interface	GTX 345
Traffic	Interface	None
TCAS II	Interface	None
GDL 60	Interface	Other LRU
GDL 69	Interface	GDL 69A
	Antenna Gain	22.00
	Cable Loss	3.0
	Calc Atten	5.0
	Automatic Muting	Off
	Mute Speed	0 KT

Interfaced Equipment		
Cross-Side Navigator	Status	Present
	Type	GTN
GDL 60	Status	Present
	Port	HSDB Port 4
GDL 69/69A	Status	Present
	Type	GDL 69A
GDL 88	Status	Not Present
ADS-B In	Status	Present
	In Source	GTX #1
11/21/2025		
file:///C:/Users/Wrightnicolas/Downloads/NXXXXX_GTN1_main_summ...		
	External Traffic Source	None
GDU	Status	Format
	GDU #1	Present GDU TXi
	GDU #2	Present GDU TXi
	GDU #3	Present GDU TXi
	GDU #4	Not Present -
GI 275	GI #1	Present
	GI #2	Not Present
	GI #3	Not Present
	GI #4	Not Present
	GI #5	Not Present
	GI #6	Not Present

✈️ Let's look at the configs on the GTN Xi's:

✈️ GTN 1:

✈️ This Unit we see GTN configured as the cross-side navigator it's important to make the distinction between GTN and GTN Xi here

✈️ We also see that this installer elected to do a redundant HSDB line while this isn't wrong it can cause issues for the GDL 60 its best to not have this here

✈️ Let's look at the configs on the GI 275

✈️ GI 1

- ✈️ This Unit we see only the GTN 650 is configured this means it wasn't looking for data from the GTN 750
- ✈️ When a GDL 60 is present all other Bluetooth access points should be turned off
 - ✈️ (exception made for GMA)
- ✈️ We also see that the GDL 60 is not configured as present that means it may have been taking databases at a slower rate

GPS 1	Interface	GTN 6XX
	Transmit Course to GTN	On
GPS 2	Interface	None
VFR GPS	Interface	Internal
Wireless Settings		
	Master	GI
	SSID	GCC Default SSID
	Password	connected
	Bluetooth Name	GI 275-48E0
	Garmin Pilot Options	Database Upload On
Airspeed Switch #1	Interface	None
	Discrete Port	None
PFD Setting Sync	Interface	HSDB
GDL 60	Interface	None
Outside Air Temperature	Interface	Other ADC
	01-MAY-25	72 minutes

Connectivity

LRU Connection Status

GDL60 1 ☒

Connectivity

Bluetooth Enable ☒

Crew Wifi ☒

Wifi Channel

Crew Wifi Airborne

Connex Remote Aircraft Status Display

Wakeup

LRU Connection Status

GDL60 1 ☒

Wakeup Configuration

Enable Wake ☒

Enable LTE AO ☒

Interval (days)

Time (UTC)

✈️ Let's look at the configs on the GDL 60:

✈️ GDL 60

- ✈️ This unit is configured properly for databases and RAS
- ✈️ Note LTE AO and Enable Wake are turned on!
- ✈️ The wake interval is set to 1 day this is allowed but means the GDL 60 wakes up everyday to look for databases this is very aggressive
- ✈️ Recommend no less than every 7 days. Ideally 15 days to catch a mid cycle update if the aircraft isn't powered on.

✈ On the GTN 750 We want you to set the GDL 60 HSDB to Other LRU. We see the GDL 60 act and I quote “strange when both HSDB lines are connected”. We can start with just config and leave the wire in there. If we still see issues, we may ask you to cap and stow or remove that wire.

- ✈ On the GDU 2 EIS display set the GPS1 and GPS 2 to the GTN 7XX and GTN 6XX We can also set the NAV to GTN 750 and GTN 650. The GPS being set will turn on the GTN and GDU 2 Communication The GTN is the master for the GDL 60 so this may be causing the issues in the packet being built. (mainly for the Flight log Offloads but can also have an effect in RAS even though it doesn't turn on)
- ✈ We also will have you configure the GI 275 for present in GDU 2 EIS display. Turns on the HSDB
- ✈ Configure the GI 275 for present in GDU 2 EIS display. This turns on the HSDB.

- ✈ The wireless in the GI 275 appears to be configured on when a GDL 60 is present it should be the only wireless access point
- ✈ GPS 1 is set for the GTN 650 only we will want to set GPS 1 for the 750 and GPS 2 for the 650. Again this turns on the HSDB communication and we want the GPS SDI to match across the whole system.
- ✈ The GDL 60 is not configured on in the GI 275



✈️ After Performing the preceding Config changes, it solved the issue



What PlaneSync™ features do you get with a WIFI only subscription?

- A. Automatic Database Download
- B. Remote Aircraft Status
- C. Automatic Data Offloading



What PlaneSync™ features do you get with a WIFI only subscription?

- A. Automatic Database Download
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What subscriptions do I need for automatic databases to work?

- A. WIFI only
- B. Iridium
- C. Garmin or Jepp database subscription
- D. Sirius XM



What subscriptions do I need for automatic databases to work?

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✈ Senior Aviation Classroom Instructor –
Maintenance

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